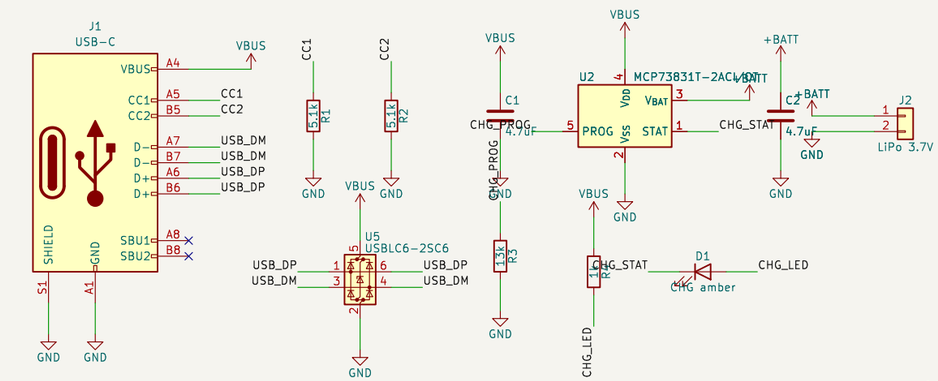
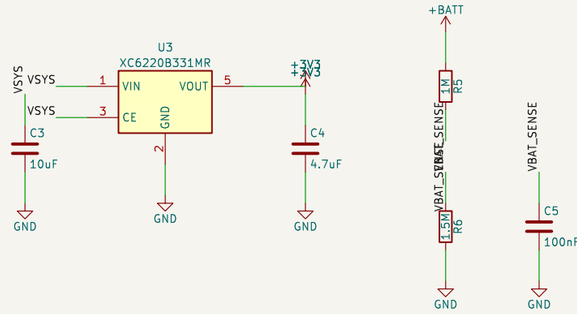
Willie Finder tracker — draft schematic rev 0.2 (not for fabrication)

USB-C, ESD, charger (77 mA), battery

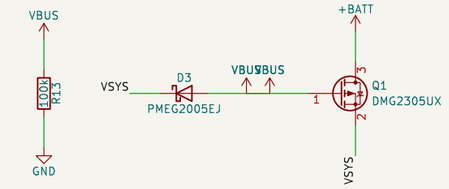


3.3 V regulator and battery sense (same divider as RAK4631 firmware expects)

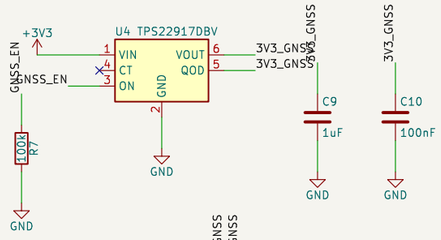




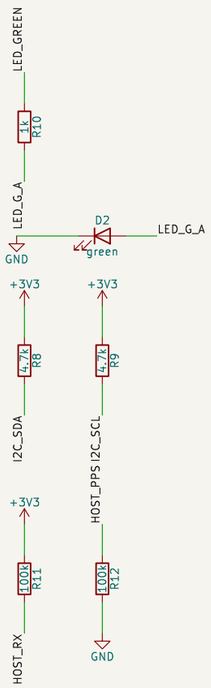
USB load sharing: on USB the system runs from VBUS, the cell only charges



GNSS on a switched rail (GNSS_EN = P1.02, high = on)

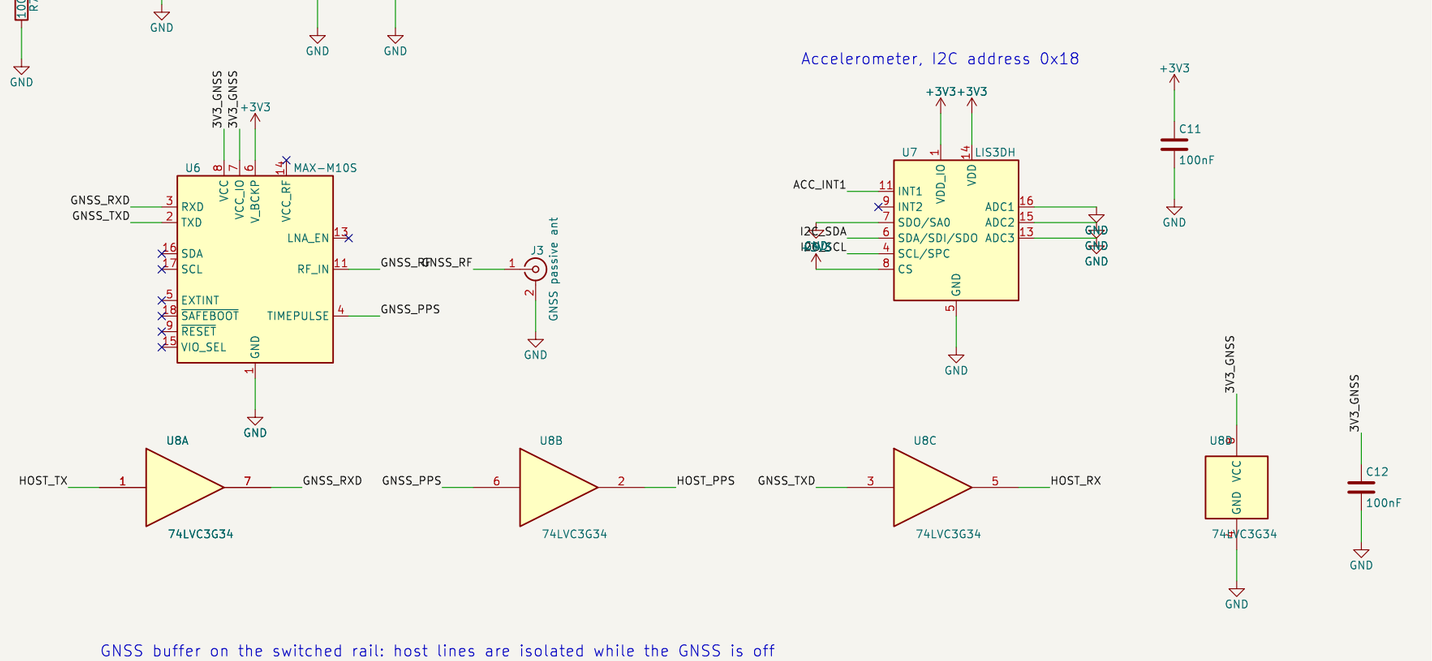


RAK4630: nRF52840 + SX1262. Pin use mirrors the Meshtastic rak4631 variant



Accelerometer, I2C address 0x18

GNSS buffer on the switched rail: host lines are isolated while the GNSS is off



LoRa and Bluetooth antennas plug into the U.FL sockets on the RAK4630 itself (−I variant).

J3 is for a passive GNSS antenna only: there is no DC bias on the coax. Route GNSS_RF as 50 ohm.

MAX-M10S VIO_SEL left open = 3.3 V I/O. Battery must have its own protection circuit.

Battery reading: divider is nominally x1.667, stock firmware uses x1.73. Calibrate against a meter.

Rev 0.2: schematic review findings 1–5 applied. Generated from pcb/gen_schematic.py.

Sheet: /
File: willie_tracker.kicad_sch

Title: Willie Finder tracker

Size: A3	Date:	**Rev: 0.2 draft**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1